MINING & GEOLOGICAL OPPORTUNITY REVIEW

Australia | Investment screening, mineral systems and exploration geography

COPPER	GOLD	LITHIUM	BROWNFIELD
Strongest stated medium-term growth	**Large, liquid market; price normalisation risk**	**Demand growth with cycle and oversupply risk**	**Infrastructure often improves project economics**

Prepared 17 October 2026 | Geographic focus: Australia-wide, with emphasis on Western Australia, South Australia, Queensland, New South Wales, Victoria, the Northern Territory and Tasmania.

Purpose and scope

Identify commercially compelling mining and exploration themes, connect them to Australian geology, and show which regions merit closer screening. This is a strategic desk study, not a reserve statement, site-specific feasibility study, securities recommendation, or legal assessment.

KEY CONCLUSION

The most defensible opportunity is usually not a greenfield discovery in isolation. It is a quality deposit in a proven mineral system, with a route to processing, power, water, logistics, approvals and contracted buyers. Geological upside should be screened together with cost position and execution risk.

1. Executive view

A national opportunity set - not a single commodity bet

Australia combines mature, infrastructure-rich mining districts with large tracts of covered geology that are incompletely tested. Geoscience Australia notes that roughly 80% of the continent is covered by younger basins and regolith, which can conceal mineral-system footprints; this is a major exploration opportunity, but also a reason discoveries are expensive and uncertain. [2]

The market backdrop is unusually mixed. The September 2026 Resources and Energy Quarterly projects copper export earnings rising, while iron ore and gold earnings ease in real terms over its outlook period. Lithium has a growing demand base but is expected to remain exposed to short-term oversupply and price moderation. These are government baseline forecasts, not guaranteed outcomes. [1]

A$422bn	A$19bn	>11% p.a.	A$3.9bn
Total resource & energy export earnings forecast for 2026-27	Copper export earnings projected by 2030-31 (real terms)	Global lithium demand growth forecast to 2031	Australia mineral exploration spend in 2025

Theme	Commercial case	Geological focus	Principal risk
Copper	Strong stated medium-term demand from electrification, grids and data centres; consider brownfield expansions and large systems. [1]	Gawler Craton; Mount Isa Inlier; Cobar Basin; Tennant Creek / Arunta.	Long lead times, capex, metallurgy, concentrate terms and water/power.
Gold	Deep liquidity, flexible scale from small underground mines to large open pits; 2025 exploration activity strongly concentrated here. [3]	Yilgarn greenstone belts; Victorian goldfields; Lachlan Fold Belt; Gawler.	Price normalisation, grade/recovery, depth, dilution and operating costs.
Lithium	Growth in batteries and storage; prefer low-cost pegmatites, expansion options and downstream/offtake alignment. [1]	Yilgarn and Pilbara LCT pegmatite provinces, especially proven camps.	Commodity cycles, conversion costs, China-linked processing and oversupply.
Iron ore	Huge established export system; most attractive for proven low-cost resources, tie-ins and productivity upgrades.	Pilbara / Hamersley Province; selected magnetite projects elsewhere.	New supply and weaker export earnings outlook; rail/port capex. [1]
Rare earths / specialty metals	Strategic value and supply-chain diversification can support project funding, but only where processing and offtake are credible. [4]	Carbonatites, alkaline intrusions, xenotime-bearing systems and heavy-mineral sands.	Separation complexity, market concentration, product qualification and limited liquidity.

Screening verdict

For a mining operator: begin with brownfield gold, copper and lithium opportunities around known deposits and infrastructure. For exploration capital: target under-cover extensions of proven mineral systems, but stage spending against geophysical, geochemical and drilling evidence. For a strategic investor: assess processing and offtake as part of the asset, not as a later add-on.

2. How Australian geology creates opportunity

Target the mineral system, not just the map location

A mineral deposit forms when a source, transport pathway, chemical/physical trap and preservation history combine. The best exploration districts are where these ingredients overlap at the right scale. Australia is geologically old, tectonically diverse and strongly weathered; ancient cratons host many of the major deposits, while younger sedimentary basins can hide older targets beneath cover. [2]

Schematic only - highlights established mineral provinces and target concepts, not land availability or mapped tenements.

Deposit model	What to look for	Australian examples / target areas
Banded iron formation (BIF) / supergene iron	Iron-rich chemical sediments, later enrichment and thick continuous ore horizons.	Hamersley Province, Pilbara, WA; Middleback Ranges, SA. [5]
Orogenic gold	Major shear zones, faults, quartz-carbonate veins, alteration and sulphide halos in greenstone/metamorphic belts.	Yilgarn Craton, WA; Victorian goldfields; parts of the Lachlan Fold Belt. [3]
Lithium-caesium-tantalum (LCT) pegmatite	Rare-element granite pegmatites, often in evolved granitic systems and near greenstone-belt margins; test mineralogy and processing.	Greenbushes / southwest WA; Pilbara and Yilgarn lithium camps. [6]
Iron oxide copper-gold (IOCG)	Large hydrothermal alteration systems, magnetic/gravity anomalies, breccias and iron oxides; ore bodies may be concealed.	Olympic Cu-Au province, Gawler Craton, SA; Tennant Creek / Arunta, NT. [7]
Sediment-hosted base metals	Favourable basin architecture, reactive horizons, faults and fluid pathways; focus on copper, lead, zinc and silver.	Mount Isa Inlier, QLD; Cobar Basin, NSW.
Palaeochannel uranium	Buried ancient river channels, permeable sands and redox fronts where oxidised uranium-bearing fluids meet reducing material.	Frome Embayment, SA; other basin/palaeochannel settings. [8]

Practical implication: the next profitable discovery may sit adjacent to an existing mine, below shallow cover, or in a poorly tested part of a known mineral province. Regional geology narrows the search; geophysics, geochemistry, drilling and metallurgy determine whether it is an economic target.

3. Precious and base metals: gold and copper

Gold - robust market depth, but costs and reserve quality decide the result

Why it screens well: gold is an internationally traded asset with multiple buyer channels, and projects range from narrow-vein underground mines to bulk-tonnage open pits. Exploration expenditure in Australia was A$3.9 billion in 2025, up 0.2% year on year; gold accounted for 40% of expenditure and gold exploration spend rose 34%. This is evidence of strong industry focus, not proof that new gold discoveries will be economic. [3]

- **High-priority geology:** extensions to known mineralised shear zones and plunging shoots in Yilgarn greenstone belts (Kalgoorlie, Leonora, Laverton, Murchison and Agnew-Wiluna districts); structurally controlled Victorian gold; and selected Lachlan Fold Belt or Gawler targets.
- **Best commercial setup:** an existing mill or nearby processing capacity, measurable resource extensions, predictable ground conditions and a mine plan with adequate grade control. Toll milling or staged underground development may reduce upfront capital in appropriate circumstances.
- **Red flags:** deep narrow lodes with high dilution, refractory ore, low recovery, high strip ratios, water/dewatering burdens, or a reserve that only works at an unusually high spot price.

Copper - strongest medium-term demand story, highest need for patient capital

The September 2026 government outlook projects Australia's copper export earnings to rise from A$14 billion in 2025-26 to A$19 billion in 2030-31 in real terms. It reports early-September copper prices above US$14,700/t and forecasts a 2027 average of US$13,570/t, easing to US$11,600/t in real terms by 2031. Demand is expected to be supported by grids, electrification, clean-energy technologies and data centres, while supply is constrained by disruptions and mine-development delays. [1]

- **Priority regions:** Gawler Craton / Olympic Cu-Au province (IOCG); Mount Isa Inlier and North West Minerals Province (base metals); Cobar Basin (structurally controlled Cu-Au and polymetallic deposits); and selected Tennant Creek / Arunta targets.
- **Geological clues:** magnetic and gravity anomalies plus alteration for IOCG; basin architecture and reactive strata for sediment-hosted copper; structural intersections and sulphide systems for Cobar-style camps. Regional geophysics is particularly important where cover is thick.
- **Commercial test:** copper grade and tonnage must be considered alongside hardness, deleterious elements, concentrate quality, recovery, water, power, tailings and distance to a smelter/export port. Large low-grade resources can still be poor investments if capital intensity is excessive.

Where incremental value often hides

Look for extensions below or beside existing deposits; improve recoveries and mine scheduling; re-process suitable legacy tailings only after independent sampling, mineralogy, environmental testing and permitting; and test satellite deposits that could feed an existing concentrator.

4. Battery and critical minerals

Lithium - attractive geology, cyclical economics

Western Australia hosts most of Australia's identified lithium resource base, mainly in hard-rock pegmatites. The strongest geological approach is to map fertile granites, pegmatite swarms and structural corridors around proven camps, then test lithium grade, spodumene content, impurities and scale. [6] The government forecasts global lithium demand to grow by more than 11% per year to 2031 and Australian mine output by about 8.8% per year. However, the market is expected to remain oversupplied in the near term; spodumene prices are forecast to moderate from around US$2,410/t in 2026 to around US$1,500/t by 2031. [1]

- **Target districts:** southwest WA around the Greenbushes system; Pilbara hard-rock lithium camps; and other Yilgarn pegmatite corridors adjacent to evolved granites and suitable host rocks.
- **Prefer:** low-cost, high-recovery spodumene projects; brownfield expansions; credible conversion partners; access to energy and reagent supply; and offtake that specifies product quality and price mechanics.
- **Avoid assuming:** a large lithium resource automatically means a viable mine. Pegmatite mineralogy, recoveries, concentrate grade, impurities, haulage and conversion route can decide project survival during downturns.

Rare earths, graphite, antimony, tungsten and vanadium - strategic upside needs processing proof

Commodity / system	Geological targets	Value-capture test
Rare earths	Carbonatite/alkaline intrusive systems, heavy-REE xenotime-style systems and monazite-bearing mineral sands; e.g. Mount Weld and Browns Range-style settings.	NdPr and heavy-REE distribution, recovery, impurity management, separation capability and long-term offtake.
Graphite	Graphite-bearing metamorphic rocks and shear-related graphite zones; prospective areas include South Australia and parts of Queensland / WA.	Flake-size distribution, purity, spherical graphite processing, qualification by anode customers and environmental footprint.
Antimony	Hydrothermal lode systems, commonly with gold; assess historic camps and untested extensions in NSW and Victoria.	Small, volatile market; secure offtake, metallurgical flowsheet and product specification before scaling.
Tin / tungsten	Granite-related greisen, vein and skarn systems; historic mineral belts in Tasmania and eastern Australia.	Grade and concentrate quality, recovery, scale and buyer access. By-products can improve economics.
Vanadium	Titanomagnetite in mafic intrusions and selected vanadium-bearing sedimentary systems.	Recoverable vanadium, co-products, processing energy, product qualification and battery-market contracts.

Strategic minerals are not automatically high-margin

A supply-chain premium can be real, but it will not rescue poor recoveries or a missing refining step. For each project, build an end-to-end model from ore through concentrate, separation/refining, product qualification, logistics and buyer contracts. Australia's Critical Minerals List is a strategic policy signal, not a guarantee of a grant, licence, price or profitable market. [4]

5. Bulk commodities and uranium

Iron ore - world-scale districts, but new entrants face a cost curve

Australia's identified iron ore resources are overwhelmingly concentrated in Western Australia, especially the Hamersley Province in the Pilbara. Many of the deposits formed in ancient banded iron formations and were upgraded by later enrichment. The major commercial advantage is scale plus integrated rail, port and operating systems. [5] The main opportunity for a new entrant is more likely to be a high-quality resource, brownfield tie-in, mine-life extension or productivity improvement than a standalone, remote greenfield mine.

Market caution: the September 2026 outlook describes softer prices amid Chinese port inventories and a well-supplied seaborne market; it projects iron ore export earnings falling from A$107 billion in 2026-27 to A$79 billion in 2030-31 in real terms as supply rises and demand growth weakens. Simandou is specifically cited as a new source of supply. [1]

Bauxite / alumina - huge deposits, with energy and processing as key constraints

The major bauxite endowments are in Cape York / Weipa in Queensland and the Darling Range in Western Australia. They are lateritic weathering profiles, so the target is commonly a broad, near-surface plateau rather than a narrow vein. Economics depends on grade and mineralogy, residue management, rehabilitation, refinery energy and caustic requirements, and access to existing alumina/aluminium infrastructure. [5]

Mineral sands - a targeted opportunity in ancient shorelines and basin margins

Heavy-mineral sands form where wave and current action concentrate dense resistant minerals in beach, dune and palaeoshoreline systems. The Murray Basin (NSW/Victoria) and Eucla Basin (WA/SA) are important exploration concepts. Assess rutile, ilmenite and zircon proportions, monazite / thorium management, ore-body continuity, water balance, rehabilitation and marketable product specifications. [5]

Uranium - promising demand outlook, highly jurisdiction-dependent

Geoscience Australia describes four broad Australian uranium deposit styles: iron-oxide breccia/IOCG, unconformity-related, sandstone-hosted and palaeochannel/calcrete. South Australia offers IOCG-associated uranium at Olympic Dam and sandstone/palaeochannel targets in the Frome Embayment; other styles occur elsewhere. [8] The government forecasts uranium prices rising from US$89/lb in 2026 to US$104/lb (real terms) in 2031 and Australian export earnings reaching about A$1.8 billion in real terms by 2030-31. [1]

Economics can be attractive where ore geometry, mining/ISR method, processing and buyer contracts fit - but state policy, approvals, radiological management, water, waste and political acceptance can dominate the schedule. A geologically attractive uranium target is not necessarily licensable.

Nickel and coal - select assets carefully

Nickel: WA hosts important komatiite-hosted sulphide systems and laterite resources, but the September 2026 outlook expects global nickel supply to outpace demand until at least 2033. Treat new capacity as high-cycle risk and prioritise low-cost, high-grade sulphides or exceptional by-product / infrastructure situations. [1]

Metallurgical coal: the Bowen and Surat basins (QLD) and Sydney-Gunnedah basins (NSW) remain established districts, with export volumes supported by steelmaking demand in India and Southeast Asia. The government forecast nevertheless sees export earnings falling in real terms from A$44 billion in 2026-27 to A$36 billion by 2030-31. Thermal coal faces a more direct demand decline. Capital, approvals and stranded-asset risk deserve heavy weighting. [1]

6. Where the geology is most compelling

Regional prospectivity matrix

Region / province	Deposit models and targets	Commercial advantage	Screening caveat
Pilbara, WA	BIF-hosted hematite / magnetite; hard-rock lithium pegmatites; selected Cu-Au and base-metal systems.	Mature mining workforce, rail/port networks, contractors and export experience.	Incumbent scale is hard to match; new iron ore must be low-cost and high-quality.
Yilgarn Craton, WA	Orogenic gold in greenstone/shear systems; komatiite-hosted Ni sulphides; LCT pegmatites.	Proven mining camps, existing mills and a deep technical ecosystem.	Mature ground can be fragmented; cover and depth increase exploration cost.
Gawler Craton, SA	IOCG Cu-Au-U, other copper-gold and selected Ni-Cu targets.	Potential for large deposits; established Olympic Dam / Prominent Hill regional knowledge.	Deep targets and cover; large capital and infrastructure requirements.
Mount Isa Inlier / NW QLD	Sediment-hosted copper, lead-zinc-silver, associated Cu-Au and regional base metals.	Historic mining district, technical skill base and multiple prospects.	Metallurgy, mine-life replacement and water / transport costs vary by site.
Cobar Basin & Lachlan Fold Belt, NSW	Structurally controlled Cu-Au / polymetallic deposits; orogenic gold and porphyry/epithermal Cu-Au targets in appropriate settings.	Existing mining services and brownfield discovery potential.	Depth, competing land uses, approvals and community / heritage engagement.
Frome Embayment, SA	Sandstone-hosted and palaeochannel uranium at permeable sand bodies / redox fronts.	Known mineral system and geophysical methods to map covered channels. [8]	Hydrogeology, recovery method, water protection and uranium policy / licensing.
Eucla & Murray basins, WA/SA/NSW/VIC	Heavy-mineral sands along palaeoshorelines and basin margins; selected uranium systems.	Potential for large, laterally continuous surficial deposits.	Water, rehabilitation, land access, product mix and by-product controls.
Tennant Creek / Arunta, NT	Ironstone-related Cu-Au, IOCG-style systems and selected under-cover targets.	Large-scale regional geology with scope to test known mineral systems beneath cover.	Remote logistics, power/water, infrastructure and native title agreements.
Cape York / Darling Range	Lateritic bauxite profiles on stable, weathered terrains.	Proven, large-volume deposits and established processing knowledge.	Refinery energy and residue, biodiversity, rehabilitation and land-access constraints.

This is a screening map of geological concepts, not a ranking of licences or a claim that unallocated ground is available. Tenement status, Indigenous land interests, conservation status, mineral rights, water and access must be checked using current state and territory data before spending money.

7. Which opportunity type fits which investor?

Four practical routes into the sector

Route	What to acquire / develop	Why it can work	Minimum evidence before committing
A. Brownfield extension	Resource extensions at producing mines; satellite deposits within trucking distance of a mill; mine-life extension.	Uses existing plant, workforce, roads, power and approvals where genuinely reusable.	Independent resource model, drilling QA/QC, metallurgy, mine plan, tailings capacity and ownership / permit review.
B. Restart / distressed asset	Care-and-maintenance mines, underused plants or historical districts with remaining mineralisation.	Potential discount to replacement cost; existing data can shorten early work.	Audit rehabilitation liabilities, equipment condition, historical resource reliability, water balance, hidden capex and restart time.
C. Advanced explorer / development asset	Resource with drilling, metallurgical testing and a preliminary study already advanced.	More geological risk retired than pure greenfield, while still offering discovery and optimisation upside.	JORC-compliant disclosure, independent technical review, recoveries, geotech, hydrology, power/water/logistics and offtake.
D. Greenfield exploration	Large land position over a proven mineral system, especially where cover conceals targets.	Highest geological upside and potential for major discoveries.	Competitive land position, mineral-system model, geophysics / geochemistry, staged drill budget and explicit stop/go criteria.

Illustrative due-diligence scorecard (use to compare assets, not to predict returns)

Criterion	Weight	Question
Geology and resource confidence	**25%**	Is mineralisation demonstrated by representative drilling and independently reviewable data? Is the resource open in a mineable direction?
Cost position and metallurgy	**20%**	Does the flowsheet deliver a saleable product at realistic recovery and cost? How does it compare to the global cost curve?
Infrastructure / logistics	**15%**	Power, water, haulage, road/rail, port, reagents, skilled labour and supply chains available on realistic terms?
Permitting, land and social licence	**15%**	Are title, approvals, native title / Indigenous agreements, heritage, biodiversity, water and closure requirements understood?
Market / offtake / processing	**15%**	Is there a credible buyer, accepted product specification and price mechanism? Are treatment/refining charges and penalties modelled?
Capital, schedule and governance	**10%**	Funding runway, contingencies, management ability, contractor availability, schedule risk and closure security are realistic?

Interpretation rule

An asset with a weaker commodity narrative but strong geology, infrastructure and low capital intensity can outperform a fashionable mineral project that requires a new mine, concentrator, chemical plant, long-distance logistics and an unproven buyer.

8. Exploration workflow for under-cover Australia

A staged plan that limits wasted drilling

Stage	Work programme	Decision gate
0. Desktop screen	Compile geological maps, mineral occurrences, historical drill logs, geochemistry, geophysics, geochronology, tenements and protected / culturally sensitive land layers. Use government pre-competitive data first.	Proceed only where deposit model, land access and title position make sense.
1. Mineral-system model	Define source, fluid pathway, trap and preservation; establish target footprints and the evidence that would falsify each model.	Select targets using explicit, testable criteria - not proximity to a famous mine alone.
2. Rapid surface screening	Map regolith and cover; use geochemistry suited to the cover type, remote sensing, field mapping and low-impact sampling where permissible.	Advance only coherent anomalies with geological context and acceptable false-positive risk.
3. Geophysics	Use magnetic, gravity, electromagnetic, radiometric or seismic methods suited to the target commodity and host rock; calibrate with known deposits.	Drill only where multiple independent datasets support a robust target.
4. First-pass drilling	Choose hole orientation and depth to test the actual geometry. Apply QA/QC, downhole surveys, independent assays and rigorous core logging.	Stop, redesign or expand based on grade, thickness, mineralogy and geological continuity.
5. Resource and metallurgy	Infill drilling, density, geotechnical and hydrological work; mineralogy, comminution, flotation / leach tests, deleterious elements and product qualification.	Move to development study only if mineability and processing are plausible.
6. Study and approvals	Scoping / PFS / DFS as appropriate; infrastructure design, closure, environmental studies, consultation, native title / Indigenous agreements, financing and binding offtake.	Investment decision after downside-price and schedule sensitivities, not on a single spot-price case.

Geophysical and geochemical targeting by commodity

Target	Useful evidence	Common trap
IOCG Cu-Au	Magnetics, gravity, alteration mapping, structural interpretation, drilling and geochemistry.	Large anomaly without economic-grade copper-gold or without feasible geometry.
Orogenic gold	Structure, alteration, vein arrays, sulphide distribution, regolith-aware geochemistry and drilling.	High nugget effect / erratic assays; shallow oxide grades do not prove underground continuity.
Pegmatite lithium	Granite fertility, pegmatite mapping, mineralogy, geochemistry and oriented drilling.	Counting all pegmatite as spodumene-bearing ore; impurities and recoveries overlooked.
Palaeochannel uranium	Airborne electromagnetics, basin architecture, channel sands, groundwater redox and radiometrics where appropriate. [8]	Channel mapping without proof of uranium grade, recoverability, hydrogeology or licensing route.

9. Economics: what makes a mine genuinely profitable?

Use a whole-of-life model, not a resource-size headline

Ore in the ground is not cash flow. A credible commercial model starts with recoverable saleable product, then deducts mining, processing, sustaining capital, freight, treatment/refining, royalties, tax, closure and financing costs. It must account for grade variability, exchange rates, diesel and power, schedule delay and commodity-price downside.

Economic driver	Why it changes profitability	Practical test
Grade and mineable geometry	Higher grade can offset haulage and processing costs; geometry drives dilution, strip ratio and recovery of the resource.	Use realistic mine shapes, dilution / mining recovery and grade-control assumptions; run low-grade and geotechnical cases.
Metallurgy and saleable product	A high in-situ grade can be worth little if recovery is low, ore is hard, deleterious elements penalise sales or product does not meet buyer specification.	Independent variability testing and locked-cycle work; confirm penalties / treatment charges with potential buyers.
Infrastructure	New power, water, rail, port, roads and camps can dominate capital cost in remote regions.	Obtain engineering estimates, capacity confirmation, tariffs, access rights and practical schedules.
Water and energy	Water scarcity, dewatering, desalination, long power connections or high diesel costs can undermine remote mine economics.	Water balance across seasons, licences and users; evaluate grid, renewables, storage and backup costs.
Approvals and closure	Heritage, biodiversity, tailings, rehabilitation and closure obligations affect schedule, financing and liabilities.	Independent closure-liability review; map every approval and agreement critical path.
Commodity cycle / contract	Spot prices, exchange rates and offtake terms determine realised price; emerging-mineral markets can be thin.	Stress-test -20% to -30% realised price, capex overrun, delayed first production and slower ramp-up.

Key macro signals from the September 2026 government outlook

Commodity	Baseline direction to 2030-31	What it means for screening
Copper	Export earnings projected A$14bn (2025-26) to A$19bn (2030-31, real).	Supports investigation of credible deposits and mine-life extensions; long project timelines remain a constraint. [1]
Gold	Export earnings projected around A$61bn in 2030-31 (real), after A$68bn in 2026-27.	Current strong price environment may support investment, but underwrite costs at lower long-term prices. [1]
Lithium	Earnings expected to peak near A$16.6bn in 2026-27, moderate to A$13.6bn in 2030-31 (real).	Demand trend is positive; avoid high-cost projects that only work near cycle peaks. [1]
Iron ore	Earnings expected to decline from A$107bn in 2026-27 to A$79bn in 2030-31 (real).	Focus on cost leadership, product quality and tie-ins, not merely size. [1]
Nickel	Market surplus expected to persist until at least 2033.	Higher hurdle rate for new projects; exceptional cost position or strategic offtake required. [1]

All forecast figures above are from the Australian Department of Industry, Science and Resources September 2026 Resources and Energy Quarterly. Figures quoted as real terms use the report's stated base where specified; earnings across different commodities should not be compared without checking the base year and scope. [1]

10. Risks, permissions and due-diligence checklist

Deal breakers to investigate before substantial capital is committed

Risk area	Questions to resolve early
Title and land access	Are exploration and mining titles valid, in good standing and correctly owned? Are overlapping claims, access agreements, private land, pastoral leases or third-party infrastructure rights involved?
Native title / Indigenous engagement	Who are the relevant Traditional Owners and representative bodies? What consultation, agreement, heritage survey and benefit-sharing processes apply? Do not treat this as a late-stage box-tick.
Environmental approvals	Are threatened species, biodiversity offsets, protected areas, cultural heritage, tailings, dust, groundwater, emissions and cumulative impacts understood? Identify Commonwealth and state processes and critical-path studies.
Water and waste	Is there sufficient lawful water supply through drought periods? What mine dewatering, tailings storage, acid/metalliferous drainage and closure obligations are expected?
Infrastructure and workforce	Are roads, rail, port slots, transmission capacity, gas / diesel, accommodation, contractors and skilled workers available at required scale and price?
Resource and metallurgy	Does the public resource comply with JORC reporting? Is drilling representative? Are recovery, comminution, impurities, geotechnical properties and hydrogeology tested across ore types?
Commercial and finance	Is offtake bankable? Have treatment/refining charges, penalties, royalties, tax, FX, hedging, working capital and closure security been modelled? Is funding sufficient through delays?
Counterparty / governance	Who controls the title and data? Are there related-party agreements, royalties, streaming or royalty agreements, litigation, historical liabilities, or unreconciled resource claims?

Suggested minimum technical data room

- Current tenement schedule, title reports, title maps, land-access and Indigenous / heritage documentation.
- All drill collars, surveys, downhole data, logs, assays, QA/QC, density, geometallurgy, geophysical data and original laboratory certificates.
- JORC-compliant resource and reserve statements, independent competent-person review and reconciliation against production where relevant.
- Metallurgical testwork, ore variability, recovery assumptions, product specifications, waste characterisation and water balance.
- Scoping / PFS / DFS studies, capex estimate class, operating-cost basis, schedule, contingency and sustaining capital.
- Independent environmental, closure, tailings, safety, rehabilitation, native title and approval-pathway assessments.
- Power, water, roads, rail, port, reagent, workforce, processing and offtake agreements or written capacity / commercial indications.

Red-line principle

Do not pay a development-stage valuation for an exploration-stage asset. Match purchase price, earn-in commitments and staged funding to the geological evidence and the approvals / engineering work actually completed.

11. Conclusion and source register

Practical priorities

- **First screen: brownfield gold, copper and lithium.** These combine known mineral systems with plausible ways to reduce risk through existing infrastructure, nearby processing and resource extensions.
- **For larger exploration upside: Gawler, Mount Isa / NW Queensland, Cobar and under-cover Yilgarn / Pilbara targets.** Use deposit-specific geophysics and staged drilling; a large anomaly is not a discovery.
- **For strategic supply chains: rare earths, graphite, antimony, tungsten and mineral sands.** Prioritise projects with defined product quality, processing routes and prospective buyers.
- **For bulk commodities: invest in cost position, not only endowment.** Pilbara iron ore and bauxite are world-class districts, but new projects need a credible advantage over incumbent producers.
- **Use conservative assumptions.** Build base, downside and severe-downside cases; explicitly price energy, diesel, capital delays, lower recovery, closure and approvals risk.

Source register

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Research date: 17 October 2026. Current commodity forecasts and policy settings can change. Geological targeting zones are regional concepts, not verified mineral resources, mine plans, or evidence of available tenements. Any real investment requires updated source checks and competent technical, legal, environmental and financial advice.